

QA Runbook — Talha Shams: Identity, Staff CRM & Operations Console

Assigned Engineer: Talha Shams

Assigned Roles & Boundaries: Keycloak IAM Gateway, OpenID Connect Authentication, Role-Based Access Control, Staff CRM Lead Inbox, Lead 360 Timeline, Opportunity Pipeline, Internal Operations & Knowledge Base

Architecture Grounding: [Blueprint §2.1 & §2.2](#), [Q014](#), [Q017](#), [Q028](#), [Q035–Q040](#), [Q054–Q060](#), [Q083](#), [Q091](#), [Q092](#)

Primary Target URLs: - Keycloak Identity Gateway: <https://identity.stackandscale.org> - Staff Portal Shell: <https://stackandscale.org/staff> - Staff Lead Inbox: <https://stackandscale.org/staff/leads> - Staff Operations Dashboard: <https://stackandscale.org/staff/operations> - Staff Knowledge & SOP Runbooks: <https://stackandscale.org/staff/knowledge> - Staff Reports & Analytics: <https://stackandscale.org/staff/reports>

1. Executive Context & System Architecture

Internal operational efficiency and customer relationship management are powered by the **Modular Staff Platform**:

- **Keycloak 26 OIDC Gateway:** Issues cryptographically signed RS256 JWT tokens, evaluated by Fastify/NestJS against the JWKS endpoint.
- **Role-Based Access Control (RBAC):** Strict boundaries separating unauthenticated visitors, standard staff, sales managers, and system administrators.
- **Relational PostgreSQL Data Core:** Houses normalized lead records, chronological timeline audits, customer organizations, and outbox event queues.

As **Talha Shams**, your mission is to verify perimeter authentication, intake the lead submitted by Muhammad Saad Khan, execute the complete CRM sales triage lifecycle, convert the prospect to a commercial opportunity, verify operations queues, and test productivity tools.

2. Testing Environment & Prerequisites

- **Browser:** Google Chrome or Firefox (in Incognito mode for initial authentication tests).
 - **Staff User Account:** Assigned to the [staff-and-scale](#) Keycloak realm with [staff](#) and [manager](#) roles.
 - **Handoff from Muhammad Saad Khan:** Ensure you have received the handoff details for [Alex Mercer](#) (alex.mercer@acmecorp-testing.com).
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3. Step-by-Step Execution Suites

Test Suite 2.1: Keycloak OIDC Authentication, Security & Denial

Context:

Per **Q028** and **Q083**, all internal [/staff](#) routes are protected by Keycloak OpenID Connect. Unauthenticated or unauthorized requests must be intercepted at the edge.

Click-by-Click Instructions:

Part A: Anonymous Access Interception (Negative Test)

1. Open a new **Incognito / Private Browsing** window.
2. In address bar, type: `https://stackandscale.org/staff` and press Enter.
3. **Observe Edge Interception:**
 - In DevTools Network tab, the request returns HTTP `302 Found`.
 - URL redirects to the Keycloak login gateway:
`https://identity.stackandscale.org/realms/stack-and-scale/protocol/openid-connect/auth? ...`
 - Verify the login card renders the official Stack & Scale brand identity. No internal data is leaked.

Part B: Invalid Credentials Handling (Negative Test)

1. In Keycloak login box, enter:
 - **Username or email:** `unauthorized-user@domain.com`
 - **Password:** `InvalidPassword999!`
2. Click “**Sign In**”.
3. **Verification:**
 - Red error alert appears: “*Invalid username or password.*”
 - Browser remains on `identity.stackandscale.org` without creating a session.

Part C: Authorized Staff Authentication (Positive Test)

1. In Keycloak login box, enter your authorized staff credentials:
 - **Username:** `staff@stackandscale.org` (or your assigned staff user)
 - **Password:** `[Your Staff Password]`
2. Click “**Sign In**”.
3. **Verify Successful Staff Shell Entry:**
 - Browser redirects back to `https://stackandscale.org/staff`.
 - In DevTools → **Application** → **Cookies**, verify secure `HttpOnly` session cookie exists.
 - In top-right header, verify your user profile avatar, name, and role badge (`Staff Operations`) are visible.

Part D: Session Termination / Sign Out

1. In top-right header, click your user dropdown → click “**Sign Out**”.
2. Verify session cookie is deleted and you are returned to login or public landing page.
3. Click browser **Back** button: verify protected staff data remains completely inaccessible.
4. Sign back in to proceed with remaining tests.

Test Suite 2.2: Lead Inbox Intake & Lead 360 Slide-Over Drawer

Context:

Per **Q014**, **Q017**, and **Q091**, inbound inquiries land in a unified triage queue where staff review lead data, attribution tags, and initiate customer contact.

Click-by-Click Instructions:

Part A: Locating the Inbound Prospect

1. In staff sidebar navigation, click on “**Leads**” (<https://stackandscale.org/staff/leads>).
2. Examine the Leads Table:
 - Columns: `Status`, `Contact Name`, `Company`, `Budget`, `Source`, `Received At`.
3. Locate the lead submitted by Muhammad Saad Khan:
 - **Contact Name:** `Alex Mercer`
 - **Company:** `Acme Global Technologies`
 - **Budget:** `$25,000 - $50,000`
 - **Status Badge:** Blue badge reading `NEW`.
 - **Received Timestamp:** Matches the time Muhammad Saad Khan executed Test 1.5.

Part B: Lead 360 Slide-Over Drawer

1. Click anywhere on the table row for `Alex Mercer`.
2. **Verify Drawer Presentation:**
 - Slide-over panel smoothly animates from the right edge.
 - Header shows: `Alex Mercer - Acme Global Technologies`.
 - Metadata section displays:
 - Email: `alex.mercer@acmecorp-testing.com` (clickable `mailto:` link).
 - Attribution: `Source: web | Form: contact_page`.
 - Inbound Message: Verbatim text entered during Test 1.5.

Part C: Lifecycle Stage Transitions (Q092)

1. In the drawer header, click the “**Lead Status**” dropdown (currently `NEW`).
2. Select `CONTACTED`.
 - Status badge turns orange `CONTACTED`.
 - Success toast appears: “*Lead status updated to Contacted.*”
 - In the **Activity Timeline** below, an audit entry appears: “*Status changed from NEW to CONTACTED by [Your Name] just now.*”
3. Click the dropdown again and select `QUALIFIED`.
 - Status badge updates to purple `QUALIFIED` and timeline logs the event.

Part D: Internal Collaboration Timeline Notes (Q055)

1. In the Lead 360 drawer, click on the “**Notes**” tab.
2. Click into the note textarea and type:

```
Completed technical discovery call with Alex Mercer.
Confirmed requirements:
1. Multi-tenant Client Portal for 20 downstream corporate accounts.
2. Private S3 file deliverables with antivirus quarantine.
3. Budget verified: $45,000 USD approved for Phase 1 execution.
```

3. Click the blue button: “**Add Note**”.
4. **Verification:**
 - Note is committed to PostgreSQL and renders at the top of the timeline.
 - Displays your author name and UTC timestamp.
 - Refresh the page: verify the note remains intact.

Test Suite 2.3: Converting Lead to Commercial Opportunity

Context:

Per **Q014** and **Q092**, qualified leads cross the boundary into active sales opportunities with defined deal value, target close date, and pipeline stage.

Click-by-Click Instructions:

1. In the Lead 360 drawer for Alex Mercer, click the primary action button: **“Convert to Opportunity”**.
 2. In the modal dialog that appears, fill in the commercial terms:
 - **Opportunity Title:** `Acme Corp – Multi-Tenant Cloud Architecture`
 - **Estimated Deal Value (\$):** `45000`
 - **Target Close Date:** Select a date 30 days in the future.
 - **Initial Pipeline Stage:** Select `Proposal / Scoping`.
 - **Assigned Account Lead:** Select your staff user.
 3. Click the green button: **“Confirm Conversion”**.
 4. **Observe the Results:**
 - Modal closes.
 - Lead status updates to `CONVERTED`.
 - An Opportunity Record ID is generated (e.g., `OPP-2026-0042`).
 - Click the link: *“View Linked Opportunity →”*.
 - Verify the Opportunity view opens, displaying the `$45,000 USD` pipeline value and `Proposal / Scoping` stage.
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Test Suite 2.4: Operations Queue Telemetry & Outbox Monitoring

Context:

Per **Q035**, **Q054**, and **Q073**, all external communications and event dispatches are processed asynchronously via PostgreSQL transactional outbox workers.

Click-by-Click Instructions:

1. In the sidebar, click **“Operations”** (`https://stackandscale.org/staff/operations`).
 2. Review the **Engine Health & Queue Dashboard**:
 - **Outbox Worker Status:** Must display `HEALTHY (Active)`.
 - **Pending Outbox Messages:** Must show `0` or low single digits.
 - **Failed / Dead-Letter Messages:** Must show `0`.
 - **Email Delivery Service:** Shows connected transactional provider (`Resend API`).
 3. Scroll down to **Recent Operations Events**:
 - Verify that your status changes and opportunity conversion events appear in the stream with unique correlation IDs.
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Test Suite 2.5: Search, Knowledge Base SOPs & Audio Feedback (SFX)

Click-by-Click Instructions:

Part A: Staff Operations Search (Q056)

1. In the top staff header, click the search bar or press `/` on your keyboard.
2. Type: `Acme`.
3. **Verification:**
 - Results group immediately into categories:
 - **Leads:** `Alex Mercer (Acme Global Technologies)`
 - **Opportunities:** `Acme Corp – Multi-Tenant Cloud Architecture`
 - Click the Opportunity result: verify direct navigation.

Part B: Staff Knowledge Base (Q059)

1. In the sidebar, click “**Knowledge**” (`https://stackandscale.org/staff/knowledge`).
2. Open the article: “*Commercial Boundary & Client Onboarding Runbook*”.
3. Verify formatted markdown sections and bash code snippets.
4. Click the “**Copy**” button on a code snippet: verify checkmark appears and snippet copies to clipboard.

Part C: Audio Feedback (SFX) Toggle

1. Locate the **Speaker / Audio Icon** in the sidebar footer.
2. Click to toggle SFX **ON**.
3. Perform an action (e.g., add a one-line test note):
 - Verify a subtle, crisp audio chime plays upon completion.
4. Click the icon again to mute.

Test Suite 2.6: Reporting & CSV Export Data Integrity

Click-by-Click Instructions:

1. In the sidebar, click “**Reports**” (`https://stackandscale.org/staff/reports`).
2. Review the live metrics cards:
 - **Total Inbound Leads:** Total count (incremented by Alex Mercer).
 - **Lead-to-Opportunity Conversion Rate:** Accurately calculated percentage.
 - **Active Pipeline Value:** Reflects the `$45,000 USD` deal.
3. Click the button: “**Export Leads to CSV**” (top-right of table).
4. Open the downloaded file `Leads-export-YYYY-MM-DD.csv`:
 - Verify column headers: `id, full_name, email, company, budget, status, created_at`.
 - Verify the row for `Alex Mercer` exists with status `CONVERTED`.

4. Handoff Protocol to Hanzala Khan (Member 3)

Copy and send this handoff block to **Hanzala Khan**:

```
=====
QA HANDOFF: TALHA SHAMS (M2) → HANZALA KHAN (M3)
=====
Timestamp: [Record UTC Date & Time]
Client Organization: Acme Global Technologies
Primary Contact: Alex Mercer (alex.mercer@acmecorp-testing.com)
Opportunity ID: [Paste Opportunity ID from Step 2.3]
Target Scope: Enterprise Cloud Migration & Private Storage
Agreed Fee: $45,000 USD

Instructions for Hanzala Khan:
Please open the Proposals Console (https://stackandscale.org/staff/proposals),
generate the formal $45,000 proposal for Acme Global Technologies, and proceed to
test the multi-tenant Client Portal, Product Accounts, and ClamAV/MinIO storage.
=====
```

5. Official Talha Shams Sign-Off Sheet

Test Case	Description	Pass / Fail	Operator Signature	Timestamp
TC-2.1	Unauthenticated <code>/staff</code> access intercepted & redirected to Keycloak	☐ Pass / ☐ Fail	_____	_____
TC-2.2	Keycloak invalid credentials error banner verified	☐ Pass / ☐ Fail	_____	_____
TC-2.3	Staff login establishes HttpOnly session & loads role-aware shell	☐ Pass / ☐ Fail	_____	_____
TC-2.4	Inbound lead appears in Lead Inbox with accurate timestamp	☐ Pass / ☐ Fail	_____	_____
TC-2.5	Lead 360 drawer renders complete contact & attribution metadata	☐ Pass / ☐ Fail	_____	_____
TC-2.6	Status transition: <code>NEW</code> → <code>CONTACTED</code> → <code>QUALIFIED</code> with audit logs	☐ Pass / ☐ Fail	_____	_____
TC-2.7	Internal timeline note saved to PostgreSQL and displayed	☐ Pass / ☐ Fail	_____	_____
TC-2.8	Lead converted to Opportunity with deal value <code>\$45,000 USD</code>	☐ Pass / ☐ Fail	_____	_____
TC-2.9	Operations queue dashboard indicates healthy outbox workers	☐ Pass / ☐ Fail	_____	_____
TC-2.10	Unified search returns indexed lead and opportunity records	☐ Pass / ☐ Fail	_____	_____
TC-2.11	Knowledge Base SOPs render with copy-to-clipboard functionality	☐ Pass / ☐ Fail	_____	_____
TC-2.12	Leads CSV export downloads with complete, uncorrupted data	☐ Pass / ☐ Fail	_____	_____